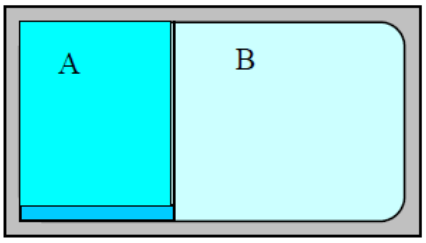
HW #2 Solutions

2.1

Solution:

C.V.: Both rooms A and B in tank.



Continuity Eq.: $m_2 = m_{A1} + m_{B1}$;

Energy Eq.: $m_2u_2 - m_{A1}u_{A1} - m_{B1}u_{B1} = {}_1Q_2 - {}_1W_2$

State 1A: (P, v) Table B.1.2, $m_{A1} = V_A/v_{A1} = 1/0.5 = 2$ kg

$$x_{A1} = \frac{v - v_f}{v_{fg}} = \frac{0.5 - 0.001061}{0.88467} = 0.564$$

$$u_{A1} = u_f + x\, u_{fg} = 504.47 + 0.564 \times 2025.02 = 1646.6 \text{ kJ/kg}$$

State 1B: Table B.1.3, $v_{B1} = 0.6173$, $u_{B1} = 2963.2$, $V_B = m_{B1}v_{B1} = 2.16\ m^3$

Process constant total volume: $V_{tot} = V_A + V_B = 3.16\ m^3$ and ${}_1W_2 = 0$

$$m_2 = m_{A1} + m_{B1} = 5.5 \text{ kg} \quad \Rightarrow \quad v_2 = V_{tot}/m_2 = 0.5746\ m^3/\text{kg}$$

State 2: T_2 , v_2 $\Rightarrow$ Table B.1.1 two-phase as $v_2 < v_g$

$$x_2 = \frac{v_2 - v_f}{v_{fg}} = \frac{0.5746 - 0.001044}{1.67185} = 0.343\ ,$$

$$u_2 = u_f + x\, u_{fg} = 418.91 + 0.343 \times 2087.58 = 1134.95 \text{ kJ/kg}$$

Heat transfer is from the energy equation

$${}_1Q_2 = m_2u_2 - m_{A1}u_{A1} - m_{B1}u_{B1} = \mathbf{-7421\ kJ}$$

2.2

Solution:

C.V. H_2O m = constant

Energy Eq.5.11: $m(e_2 - e_1) = m(u_2 - u_1) = {}_1Q_2 - {}_1W_2$

Process : P = constant (forces on piston constant)

$$\Rightarrow {}_1W_2 = \int P\, dV = P_1 (V_2 - V_1)$$

Properties from Table B.1.1

State 1: $v_1 = 0.1/50 = 0.002\ m^3/kg$ => 2-phase as $v_1 < v_g$

$$x_1 = \frac{v_1 - v_f}{v_{fg}} = \frac{0.002 - 0.001061}{0.88467} = 0.001061$$

$$h_1 = 504.68 + 0.001061 \times 2201.96 = 507.02\ kJ/kg$$

State 2: $v_2 = 0.5/50 = 0.01\ m^3/kg$ also 2-phase same P

$$x_2 = \frac{v_2 - v_f}{v_{fg}} = \frac{0.01 - 0.001061}{0.88467} = 0.01010$$

$$h_2 = 504.68 + 0.01010 \times 2201.96 = 526.92\ kJ/kg$$

Find the heat transfer from the energy equation as

$${}_1Q_2 = m(u_2 - u_1) + {}_1W_2 = m(h_2 - h_1)$$

$${}_1Q_2 = 50\ kg \times (526.92 - 507.02)\ kJ/kg = \mathbf{995\ kJ}$$

[Notice that ${}_1W_2 = P_1 (V_2 - V_1) = 200 \times (0.5 - 0.1) = 80\ kJ$]

2.3

C.V. A + B Force balance on piston: $P_A A = P_B A$

So the final state in A and B is the same.

State 1A: Table A.7 $u_{A1} = 214.364$ kJ/kg,

$m_A = P_{A1}V_{A1}/RT_{A1} = 200 \times 1/(0.287 \times 300) =$ **2.323 kg**

State 1B: Table A.7 $u_{B1} = 759.189$ kJ/kg,

$m_B = P_{B1}V_{B1}/RT_{B1} = 1000 \times 1/(0.287 \times 1000) =$ **3.484 kg**

For chosen C.V. ${}_1Q_2 = 0$, ${}_1W_2 = 0$ so the energy equation becomes

$$m_A(u_2 - u_1)_A + m_B(u_2 - u_1)_B = 0$$

$$(m_A + m_B)u_2 = m_A u_{A1} + m_B u_{B1}$$

$$= 2.323 \times 214.364 + 3.484 \times 759.189 = 3143 \text{ kJ}$$

$$u_2 = 3143/(3.484 + 2.323) = 541.24 \text{ kJ/kg}$$

From interpolation in Table A.7: $\Rightarrow$ $\mathbf{T_2 = 736\ K}$

$$P = (m_A + m_B)RT_2/V_{tot} = 5.807 \text{ kg} \times 0.287 \frac{\text{kJ}}{\text{kg K}} \times 736 \text{ K}/ 2 \text{ m}^3 = \mathbf{613\ kPa}$$

2.4

Solution:

Energy Eq.3.5: $m(u_2 - u_1) = {}_1Q_2 - {}_1W_2$

Process: $P = C$; ${}_1W_2 = \int PdV = P(V_2 - V_1)$

Table B.1.2

State 1: $v_1 = 0.46246\ m^3/kg$; $u_1 = 2553.55\ kJ/kg$

State 2: $v_2 = v_1/2 = 0.23123\ m^3/kg = v_f + x\ v_{fg}$;

$$x_2 = \frac{v_2 - v_{f2}}{v_{fg2}} = \frac{0.23123 - 0.001084}{0.46138} = 0.4988$$

$$u_2 = u_{f2} + x_2\ u_{fg2} = 604.29 + x_2 \times 1949.26 = 1576.58\ kJ/kg$$

Process: $P = C$ so the work term integral is

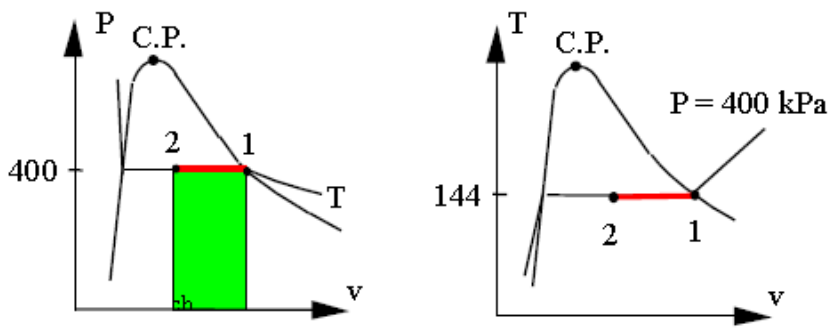
$${}_1W_2 = P(V_2 - V_1) = 400\ kPa \times 0.2 \times (0.23123 - 0.46246)\ m^3$$

$$= \mathbf{-18.5\ kJ}$$

From the energy equation

$${}_1Q_2 = m(u_2 - u_1) + {}_1W_2 = 0.2\ (1576.58 - 2553.55) - 18.5$$

$$= \mathbf{-213.9\ kJ}$$



2.5

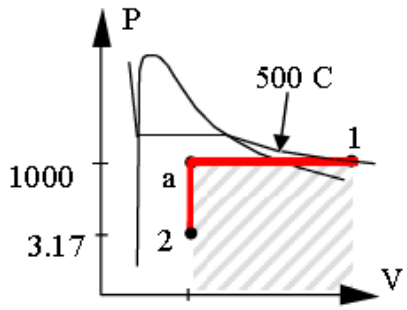
Energy Eq.: $m(u_2 - u_1) = {}_1Q_2 - {}_1W_2$

Process Eq: $P = C$ if $v > v_{stop}$; $V = C$ if $P < P_{float}$

State 1: $v_1 = 0.35411\ m^3/kg$, $u_1 = 3124.34\ kJ/kg$

State a: $v_a = v_1/2 = 0.177055\ m^3/kg < v_{g\ 1000\ kPa}$ so $T_a = T_{sat\ 1000\ kPa} = 179.9°C$

The possible state 2 (P,V) combinations are shown. State "a" is (1000 kPa, v_a) so it is two-phase with $T_a = 180°C > T_2$



$P_2 = P_{sat\ 25\ C} = 3.169\ kPa$ and $v_2 = v_a$

$x_2 = (v_2 - v_f)/v_{fg} = (0.177 - 0.001003)/43.358$

$= 0.0040604$

$u_2 = u_f + x_2\, u_{fg} = 104.86 + x_2\, 2304.9 = 114.219\ kJ/kg$

${}_1W_2 = m \int P\, dv = m\, P_1\, (v_2 - v_1)$ [see area below process curve in figure]

$= 0.1\ kg \times 1000\ kPa \times (0.177055 - 0.35411)\ m^3/kg = -17.706\ kJ$

${}_1Q_2 = m(u_2 - u_1) + {}_1W_2 = 0.1\ kg\ (114.219 - 3124.34)\ kJ/kg - 17.706\ kJ$

$= -318.72\ kJ$

2.6

Solution:

C.V. R-134a, this is a control mass.

Continuity: $m_2 = m_1 = m$;

Energy Eq.3.5: $m(u_2 - u_1) = {}_1Q_2 - {}_1W_2 = -500 \text{ kJ} - {}_1W_2$

State 1: T_1 , P_1 Table B.5.2, $v_1 = 0.04226 \text{ m}^3/\text{kg}$; $u_1 = 390.52 \text{ kJ/kg}$

$\Rightarrow V_1 = mv_1 = \mathbf{0.211 \text{ m}^3}$

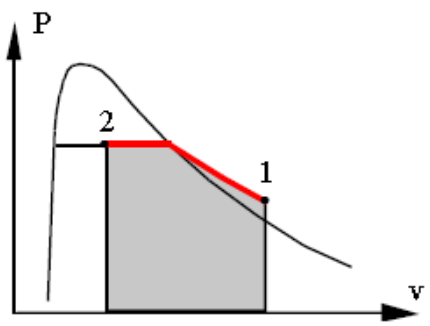
State 2: T_2 , x_2 $\Rightarrow$ Table B.5.1

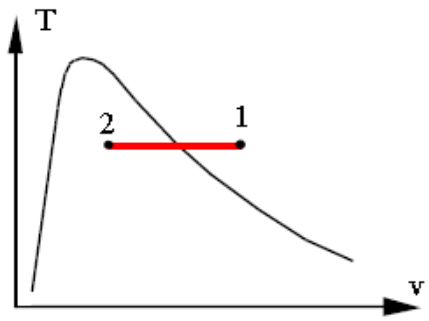
$u_2 = 227.03 + 0.5 \times 162.16 = 308.11 \text{ kJ/kg}$,

$v_2 = 0.000817 + 0.5 \times 0.03524 = 0.018437 \text{ m}^3/\text{kg} \Rightarrow V_2 = mv_2 = \mathbf{0.0922 \text{ m}^3}$

${}_1W_2 = -500 \text{ kJ} - m(u_2 - u_1) = -500 \text{ kJ} - 5 \text{ kg} \times (308.11 - 390.52) \text{ kJ/kg}$

$= \mathbf{-87.9 \text{ kJ}}$





2.7

Solution:

C.V. Total tank. Control mass of constant volume.

Mass and volume: $m_2 = m_A + m_B$; $V = V_A + V_B = 1.5\ m^3$

Energy Eq.: $U_2 - U_1 = m_2 u_2 - m_A u_{A1} - m_B u_{B1} = Q - W = 0$

Process Eq.: $V = \text{constant} \Rightarrow W = 0$; Insulated $\Rightarrow Q = 0$

Ideal gas at 1: $m_A = P_{A1}V_A/RT_{A1} = 250 \times 0.5/(0.287 \times 300) = 1.452$ kg

$u_{A1} = 214.364$ kJ/kg from Table A.7

Ideal gas at 2: $m_B = P_{B1}V_B/RT_{B1} = 500 \times 1/(0.287 \times 1000) = 1.742$ kg

$u_{B1} = 759.189$ kJ/kg from Table A.7

$m_2 = m_A + m_B = 3.194$ kg

$$u_2 = \frac{m_A u_{A1} + m_B u_{B1}}{m_2} = \frac{1.452 \times 214.364 + 1.742 \times 759.189}{3.194} = 511.51 \text{ kJ/kg}$$

=> Table A.7.1: $\mathbf{T_2 = 698.6\ K}$

$P_2 = m_2 RT_2/V = 3.194 \times 0.287 \times 698.6/1.5 =$ **426.9 kPa**

2.8

Solution:

Using constant specific heat:

$\Delta h = C_{po}\Delta T = 0.842\ (1500 - 30) =$ **1237.7 kJ/kg**

Using Table A.8:

$$30°C = 303.15\ K \Rightarrow h = 214.38 + \frac{3.15}{50}(257.9 - 214.38) = 217.12 \text{ kJ/kg}$$

$1500°C = 1773.15\ K \Rightarrow$

$$h = 1882.43 + \frac{73.15}{100}(2017.67 - 1882.43) = 1981.36 \text{ kJ/kg}$$

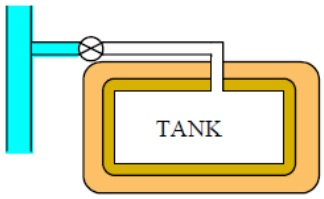
$\Delta h = 1981.36 - 217.12 =$ **1764.2 kJ/kg**

The result from A.8 is best. For large ΔT or small ΔT at high T_{avg}, constant specific heat is poor approximation or it must be evaluated at a higher T (A.5 is at 25^0C).

2.9

Solution:

C.V. Tank:



Continuity Eq.6.15: $m_2 = m_i$

Energy Eq.6.16: $m_2 u_2 = m_i h_i$

Table A.7: $u_2 = h_i = 293.64$ kJ/kg

$$\Rightarrow T_2 = \mathbf{410.0\ K}$$

$$m_2 = \frac{P_2 V}{R T_2} = \frac{600 \times 0.025}{0.287 \times 410} = \mathbf{0.1275\ kg}$$

Assuming constant specific heat,

$$h_i = u_i + RT_i = u_2\ , \quad RT_i = u_2 - u_i = C_{VO}(T_2 - T_i)$$

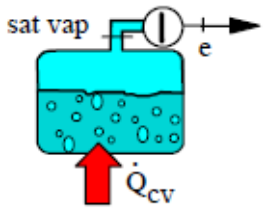
$$C_{VO}T_2 = (C_{VO} + R)T_i = C_{PO}T_i\ , \quad T_2 = \left(\frac{C_{PO}}{C_{VO}}\right) T_i = kT_i$$

For $T_i = 293.2$K & constant C_{PO}, $T_2 = 1.40 \times 293.2 = \mathbf{410.5\ K}$

2.10

Solution:

C.V. Tank, no work but heat transfer in and flow out. Denoting State 1: initial state, State 2: valve opens, State 3: final state.



Continuity Eq.: $m_3 - m_1 = -m_e$

Energy Eq.: $m_3u_3 - m_1u_1 = -m_eh_e + {}_1Q_3$

State 1 Table B.1.2: $v_1 = v_f + x_1v_{fg} = 0.001043 + 0.01\times1.69296$

$= 0.01797\ m^3/kg$

$u_1 = u_f + x_1u_{fg} = 417.33 + 0.01\times2088.72 = 438.22\ kJ/kg$

$m_1 = V/v_1 = 0.2\ m^3/(0.01797\ m^3/kg) = 11.13\ kg$

State 3 (2MPa): $v_3 = v_f + x_3v_{fg} = 0.001177 + 0.9\times0.09845 = 0.8978\ m^3/kg$

$u_3 = u_f + x_3u_{fg} = 906.42 + 0.9\times1693.84 = 2430.88\ kJ/kg$

$m_3 = V/v_3 = 0.2\ m^3/(0.08978\ m^3/kg) = 2.23\ kg$

Exit state (2MPa): $h_e = h_g = 2799.51\ kJ/kg$

Hence: $m_e = m_1 - m_3 = 11.13\ kg - 2.23\ kg = \mathbf{8.90\ kg}$

Applying the 1st law between state 1 and state 3

$${}_1Q_3 = m_3u_3 - m_1u_1 + m_eh_e$$

$$= 2.23 \times 2430.88 - 11.13 \times 438.22 + 8.90 \times 2799.51$$

$$= 25\ 459\ kJ = \mathbf{25.46\ MJ}$$

2.11

Solution:
The compressor, turbine and nozzle are all steady state single flow devices and they are adiabatic.

We will use air properties from table A.7.1:

$h_1 = 260.32$, $h_2 = 800.28$, $h_3 = 1635.80$, $h_4 = 933.15$, $h_5 = 649.53$ kJ/kg

Energy equation for the compressor gives

$$w_{c\ in} = h_2 - h_1 = 800.28 - 260.32 = \mathbf{539.36\ kJ/kg}$$

Energy equation for the turbine gives

$$w_T = h_3 - h_4 = 1635.80 - 933.15 = \mathbf{702.65\ kJ/kg}$$

Energy equation for the nozzle gives

$$h_4 = h_5 + \tfrac{1}{2} V_5^2$$

$$\tfrac{1}{2} V_5^2 = h_4 - h_5 = 933.15 - 649.53 = 283.62 \text{ kJ/kg}$$

$$V_5 = [2(h_4 - h_5)]^{1/2} = (2 \times 283.62 \times 1000)^{1/2} = \mathbf{753\ m/s}$$

2.12

Solution:

Continuity Eq.6.3: $\dot{m}_i = A_i \mathbf{V}_i / v_i = \dot{m}_e = A_e \mathbf{V}_e / v_e$,

Energy Eq.(per unit mass flow)6.13: $h_i + \frac{1}{2}\mathbf{V}_i^2 = h_e + \frac{1}{2}\mathbf{V}_e^2$

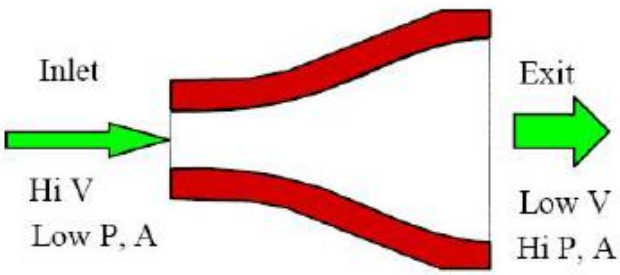
$$h_e - h_i = \frac{1}{2} \times 200^2/1000 - \frac{1}{2} \times 20^2/1000 = 19.8 \text{ kJ/kg}$$

$$T_e = T_i + (h_e - h_i)/C_p = 300 + 19.8/1.004 = \mathbf{319.72\ K}$$

Now use the continuity equation and the ideal gas law

$$v_e = v_i \left(\frac{A_e \mathbf{V}_e}{A_i \mathbf{V}_i}\right) = (RT_i/P_i)\left(\frac{A_e \mathbf{V}_e}{A_i \mathbf{V}_i}\right) = RT_e/P_e$$

$$P_e = P_i \left(\frac{T_e}{T_i}\right)\left(\frac{A_i \mathbf{V}_i}{A_e \mathbf{V}_e}\right) = 100\left(\frac{319.72}{300}\right)\left(\frac{100 \times 200}{860 \times 20}\right) = \mathbf{123.92\ kPa}$$



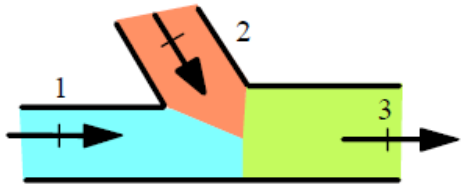
2.13

Solution:

Cont. $\dot{m}_i = \dot{m}_e = \dot{m}$

Energy $\dot{m}_1 h_1 + \dot{m}_2 h_2 = \dot{m}_3 h_3$

$= (\dot{m}_1 + \dot{m}_2) h_3$



Mixing section

$$\dot{m}_1 (h_3 - h_1) + \dot{m}_2 (h_3 - h_2) = 0$$

$$\dot{m}_1 C_p (T_3 - T_1) + \dot{m}_2 C_p (T_3 - T_2) = 0$$

$$T_3 = (\dot{m}_i/\dot{m}_3)/T_1 + (\dot{m}_2/\dot{m}_3)T_2$$

We need to find the mass flow rates

$$v_1 = RT_1/P_1 = (0.287 \times 293)/100 = 0.8409 \text{ m}^3/\text{kg}$$

$$v_2 = RT_2/P_2 = (0.287 \times 473)/100 = 1.3575 \text{ m}^3/\text{kg}$$

$$\dot{m}_1 = \frac{\dot{V}_1}{v_1} = \frac{1}{0.8409} = 1.1892 \frac{\text{kg}}{\text{s}}, \quad \dot{m}_2 = \frac{\dot{V}_2}{v_2} = \frac{2}{1.3575} = 1.4733 \frac{\text{kg}}{\text{s}}$$

$$\dot{m}_3 = \dot{m}_1 + \dot{m}_2 = 2.6625 \text{ kg/s}$$

$$T_3 = \frac{1.1892}{2.6625} \times 20 + \frac{1.4733}{2.6625} \times 200 = \mathbf{119.6^o\ C}$$

$$v_3 = \frac{RT_3}{P_3} = \frac{0.287\ (119.6 + 273)}{100} = 1.1268 \text{ m}^3/\text{kg}$$

$$\dot{V}_3 = \dot{m}_3\ v_3 = 2.6625 \times 1.1268 = \mathbf{3.0\ m^3/s}$$

2.14

Solution:

C.V. Mixing Section

Continuity Eq.6.9: $\dot{m}_1 + \dot{m}_2 = \dot{m}_3$ => $\dot{m}_3 = 2 + 1.5 = 3.5$ kg/s

Energy Eq.6.10: $\dot{m}_1 h_1 + \dot{m}_2 h_2 = \dot{m}_3 h_3$

$$h_3 = (\dot{m}_1 h_1 + \dot{m}_2 h_2) / \dot{m}_3 \;;$$

For a constant specific heat divide the equation for h_3 with C_p to get

$$T_3 = \frac{\dot{m}_1}{\dot{m}_3} T_1 + \frac{\dot{m}_2}{\dot{m}_3} T_2 = \frac{2}{3.5} 2000 + \frac{1.5}{3.5} 500 = \mathbf{1357\ K}$$

2.15

Solution:

C.V. Mixing Section

Continuity Eq.6.9: $\dot{m}_1 + \dot{m}_2 = \dot{m}_3$ => $\dot{m}_3 = 2 + 1.5 = 3.5$ kg/s

Energy Eq.6.10: $\dot{m}_1 h_1 + \dot{m}_2 h_2 = \dot{m}_3 h_3$

$h_3 = (\dot{m}_1 h_1 + \dot{m}_2 h_2) / \dot{m}_3$;

Using A.7 we look up the h at states 1 and 2 to calculate h_3

$$h_3 = \frac{\dot{m}_1}{\dot{m}_3} h_1 + \frac{\dot{m}_2}{\dot{m}_3} h_2 = \frac{2}{3.5} 2251.58 + \frac{1.5}{3.5} 503.36 = 1502 \text{ kJ/kg}$$

Now we can backinterpolate to find at what temperature do we have that h

$$T_3 = 1350 + 50 \frac{1502 - 1455.43}{1515.27 - 1455.43} = \mathbf{1389\ K}$$

This procedure is the most accurate.

2.16

Solution:

C.V. Diffuser, Steady state, 1 inlet, 1 exit flow, no q, no w.

Continuity Eq.4.3: $\dot{m}_i = \dot{m}_e = (AV/v)$

Energy Eq.4.12: $\dot{m}\,(h_i + \frac{1}{2}V_i^2) = \dot{m}\,(\frac{1}{2}V_e^2 + h_e)$

$$h_e - h_i = C_p\,(T_e - T_i) = \frac{1}{2}V_i^2 - \frac{1}{2}V_e^2 = \frac{1}{2}\left(\frac{900 \times 1000}{3600}\right)^2 - \frac{1}{2}(80)^2$$

$$= \frac{1}{2}(250)^2 - \frac{1}{2}(80)^2 = 28050 \text{ J/kg} = 28.05 \text{ kJ/kg}$$

$\Delta T = 28.05/1.004 = 27.9 \quad \Rightarrow \qquad T_e = -5 + 27.9 = \mathbf{22.9°C}$

Now use the continuity eq.:

$$A_iV_i/v_i = A_eV_e/v_e \quad \Rightarrow \qquad v_e = v_i\left(\frac{A_eV_e}{A_iV_i}\right)$$

$$v_e = v_i \times \frac{1.2 \times 80}{1 \times 250} = v_i \times 0.384$$

Ideal gas: $Pv = RT \quad \Rightarrow \quad v_e = RT_e/P_e = RT_i \times 0.384/P_i$

$P_e = P_i\,(T_e/T_i)/0.384 = 50 \text{ kPa} \times 296/(268 \times 0.384) = \mathbf{143.8\ kPa}$